Human-centered design

Privacy/Behavior Analysis in Mixed Reality

- **Background**: Head-mounted mixed-reality devices can render and overlay realistic 3D content onto the physical environment, enabling a wealth of interactive possibilities. These devices are equipped with a rich set of sensors that collect personal and sensitive information (e.g., body motion, eye gaze, hand joints, and facial expression), which can be used to uniquely identify a user, even without explicit identifiers, and make inferences about human behavior
- **Goal**: Find useful and relevant datasets, preprocess them, and perform privacy/behavior analysis
- **Deliverables**:
  - Find existing datasets for user pose, gaze and gestures
  - Perform privacy/behavior analysis on the data based on existing literature
- **Software:** Python, Unity
- **References**:
  - MURMR: A Multimodal Sensing Framework for Automated Group Behavior Analysis in Mixed Reality, 2026
  - BehaVR: User Identification Based on VR Sensor Data, 2023
  - Understanding User Identification in Virtual Reality Through Behavioral Biometrics and the Effect of Body Normalization, 2021
  - Personal Identifiability of User Tracking Data During Observation of 360-degree VR Video, 2020
- **Team Size: 3**

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Project Motivation: We have plenty of hardware experience both from the CompE core curriculum and from other classes such as 390B where Luke created an air quality sensing system and Aidan created a sun tracking solar charger in our respective teams. Furthermore, Aidan has experience with a data analysis project Aidan did sophomore year with the guidance of a grad student in Khwarizmi lab that has given me background knowledge in working with and processing large datasets in Python. My senior design project will also involve computer vision, user tracking, and gesture control which makes me especially interested in focusing on this project which will hopefully go hand in hand with my SDP.

Design Goals:

1. Measure whether users can be identified from head motion, controller motion, and eye gaze data separately and in combination
2. Identify which sensor signals contribute most to identification
3. Test simple privacy measures like lower sampling rates and reduced coordinate precision and measure how they affect identification accuracy and basic behavior measures
4. Produce a reproducible analysis with clear charts limitations and recommendations for XR data collection

Deliverables: “Find existing datasets for user pose, gaze and gestures”, “Perform privacy/behavior analysis on the data based on existing literature”

System Blocks: Public dataset -> Data checks and preprocessing -> Time window features -> Identification models -> Privacy transformations -> Evaluation and visualization

- Data checks and preprocessing: Inspect available users and sessions, synchronize signals, handle missing values, and divide recordings into short time windows.
- Time window features: Calculate motion features such as speed and rotation, gaze features such as direction changes. Keep each sensor group separate for comparison.
- Identification models: Train a simple baseline classifier, then compare head motion, controllers, haze and combined signals. Where the data allows, train on one session and test on a different session from the same users.
- Privacy analysis: Reduce sampling rate and/or coordinate precision and rerun the models.
- Evaluations: Report the identification accuracy and macro F1 score, including any changes to behavior measures like movement speed and gaze direction. We will include a summary of the privacy utility trade off charts.

Hardware and software requirements:

- Hardware: Two computers capable of processing CSV files; approximately 16GB Ram and sufficient storage for the selected recordings.
- Needed software: Python, Jupyter, pandas, NumPy, scikit-learn, and Matplotlib or Seaborn.
- Optional software: Unity to be used for a visual replay or demo if we have enough time.

Team responsibilities:

- 1: Dataset review, data cleaning, synchronization, feature extraction, and documentation of data limitations.
- 2: Identification models, privacy transformations metrics, and result visualizations.
- Both: We will choose the evaluation protocol, review each other's results, write the report, and we will present the findings.

Timelines:

1. Review related material and confirm dataset access and license, define research questions
2. Inspect signals, users, sessions, and missing data; finalize the evaluation plan
3. Build preprocessing and time window feature pipeline
4. Train baseline identification models and compare sensor groups
5. Apply sampling rate and precision reductions
6. Evaluate identification risk and behavior-measure changes and create charts
7. Review results, limitations, and reproducibility, and draft the report and presentation
8. Finalize report, code, figures, and presentation

Expected deliverables: a dataset and literature review, documented Python analysis, comparison charts, a privacy utility discussion and a final presentation